

Dr. Aayush Garg

Nationality: Luxembourgish (EU citizen)

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Research Interests

Cybersecurity, Vulnerability Analysis, Prediction, and Augmentation, Quality Assurance, AI-driven Software Security, Mutation Testing, Machine Learning, Large Language Models

Education

Ph.D. in Computer Science

University of Luxembourg, Luxembourg

July 2019 – May 2023

Dissertation: *Guiding Quality Assurance Through Context Aware Learning*

Advisor: Prof. Yves Le Traon; Co-advisor: Prof. Mike Papadakis

M.S. in Computer Science

Boston University, USA

Concentration: *Security*

August 2017 – January 2019

GPA: 3.8/4.0

B.Tech. in Computer Science

Amity University, India

April 2006 – March 2010

GPA: 7.0/10.0

Grant Writing

ROSETTA Postdoctoral Fellowship

2026

Marie Skłodowska-Curie Actions (MSCA) COFUND

Above funding threshold score: 0.8/1.0

Research Experience

Scientist (Research & Technology)

Luxembourg Institute of Science and Technology

March 2024 – Present

Luxembourg

- Developing AI-guided methods to improve software security, focusing on real (existing) and synthetically generated vulnerabilities.
- Exploring LLMs for automated patch generation, software self-healing, and improved developer experience through code-assistance tools.
- Developing automated solutions to identify attack vectors in telecom core networks.
- Utilizing Deep Learning, particularly Transformer architectures, to identify API fuzzing attack patterns.
- Developing automated assurance techniques to ensure the quality of Generative AI-enabled educational activities.
- Designing frameworks to transform cyber threat intelligence into executable cyber range scenarios for providing hands-on security training to industry professionals.

Doctoral Researcher Interdisciplinary Centre for Security, Reliability and Trust (SnT)
July 2019 – March 2024 University of Luxembourg

- Designed AI-driven solutions to automate feature extraction from source code, replacing manual efforts.
- Trained Neural Networks achieving up to 87% Precision and Recall in task-specific classifications.
- Employed LLMs and GPTs to synthetically generate and induce artificial faults, increasing test effectiveness by up to 65%.

Teaching Experience

Teaching Assistant Faculty of Science, Technology and Medicine (FSTM)
July 2019 – May 2023 University of Luxembourg

- Integrated real-world case studies into coursework to simulate business-oriented applications of AI and information systems.
- Conducted lectures on “Introduction to Machine and Deep Learning” for Bachelors in Computer Science(BICS), mentoring students in AI-driven software development and QA.
- Delivered lectures on “Introduction to Software Testing” for Bachelors in Applied Information Technology (BINFO), guiding students in secure coding methodologies.
- Assisted in inclusive course material creation to enhance student engagement.
- Recorded lecture series available [online](#) to support students during COVID-19 lockdown.

Industry Experience

Senior Software Engineer Futures First, India
September 2014 – February 2017

- Developed multi-threaded Windows applications to process and display stock prices and trade updates with a 3-second refresh rate.
- Implemented microservices to capture up to 12 real-time stock price updates per second via market APIs.
- Engineered a Futures and Commodities trading platform, enabling a minimum of 95% of organizational market investments.

Senior Software Engineer Indus Valley Partners, India
August 2012 – September 2014

- Built investment-compliance applications to automate legal due diligence auditing, reducing manual effort by up to 85%.
- Implemented microservices to enable configurable email alerts, streamlining at least 80% of business process workflows.
- Integrated portfolio dashboards and reporting capabilities for 11 Business Experts (SMEs), facilitating comprehensive debt investment overviews.

Software Engineer Fiserv, India
May 2010 – August 2012

- Crafted web applications to streamline investment portfolio metrics capturing, resulting in a 53% increase in efficiency.

- Resolved defects and maintained source code for three large-scale Electronic Fund Transfer banking projects.
- Ensured high-quality (minimum 85% defect-free) applications through peer-reviewing code modifications.

Causal Analysis and Resolution Coordinator

Fiserv, India

May 2010 – August 2012

- Performed defect root cause analysis and processed inefficiencies for 95% of Business Unit.
- Executed corrective actions to maintain 85% defect-free software quality and to achieve 15% increased module delivery efficiency.

Skills

- **Programming Languages:** Python, Java, C++, C#
- **AI/ML Frameworks & Libraries:** PyTorch, Pandas, TensorFlow, Keras, Scikit-learn, Hugging Face
- **Techniques:** Machine Learning, Deep Learning, Natural Language Processing, Prompt Engineering, Static and Dynamic Analysis
- **Security and Testing:** Vulnerability Assessment and Prediction, Artificial Vulnerability Generation, Vulnerability Injection, Mutation Testing
- **NLP and LLMs:** Large Language Models (LLM), Generative Pretrained Transformers (GPT), Neural Machine Translation (NMT), Encoder-Decoders, Transformers
- **Tools and Platforms:** Git, Docker, SQL Server, Apache Cassandra, RabbitMQ

Professional Activities

Conference and Workshop Roles

- **Track Chair and Organizing Committee Member:**
 - International Conference on Evaluation and Assessment in Software Engineering (EASE) 2025, Learnings/Reflections of Evaluation and Assessment Projects in Software Engineering (Learnings & Reflections) Track.
- **Program Committee Member:**
 - ACM International Conference on the Foundations of Software Engineering (FSE) 2027, Industry Track.
 - IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026, Industry Showcase Track.
 - IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026, Tools and Datasets Track.
 - IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026, New Ideas and Emerging Results (NIER) Track.
 - IEEE International Conference on Software Testing, Verification and Validation (ICST) 2026, Mutation 2026 Workshop.
 - International Conference on Software Maintenance and Evolution (ICSME) 2026, Industry Track
 - International Conference on Software Maintenance and Evolution (ICSME) 2026, Visions and Emerging Results Track.
 - International Symposium on Information and Communication Technology (SOICT) 2026, Software Engineering, Trusted Digital Platforms, and Smart Services Track.
 - IEEE/ACM International Conference on Automated Software Engineering (ASE) 2025, New Ideas and Emerging Results (NIER) Track.

- International Symposium on Software Testing and Analysis (ISSTA) 2025, Tool Demonstrations Track.
- Pacific Rim International Conference on Artificial Intelligence (PRICAI) 2025, Research Track.
- IEEE/ACM International Conference on Automated Software Engineering (ASE) 2024, NIER Track.
- IEEE International Conference on Software Testing, Verification and Validation (ICST) 2024, Mutation 2024 Workshop.

Peer Reviewing (Journals)

- Springer Nature Empirical Software Engineering (EMSE) Journal, since October 2025
- ACM Transactions on Software Engineering and Methodology (TOSEM) Journal, since November 2024.
- Springer International Journal of Machine Learning and Cybernetics, since October 2024.
- Springer Automated Software Engineering Journal, since September 2024.
- Springer International Journal of Information Security, since September 2024.
- Springer Scientific Reports, since May 2024.
- Elsevier Computers & Security Journal, since January 2024.
- Springer Software Quality Journal, since December 2023.
- Software Testing, Verification and Reliability (STVR) Journal, since December 2023.
- IEEE Transactions on Software Engineering (TSE) Journal, since August 2022.

Invited Talks and Presentations

- **ICST 2024**, Toronto, Canada: Presented “On the Coupling between Vulnerabilities and LLM-generated Mutants: A Study on Vul4J dataset,” May 30, 2024.
- **ISSRE 2023**, Florence, Italy: Presented “Enabling Efficient Assertion Inference,” October 12, 2023.
- **CREST, University of Adelaide**, Australia: Delivered guest lectures on “Guiding Quality Assurance Through Context Aware Learning,” August 2023.
- **ICSE 2023**, Melbourne, Australia: Presented “Learning from What We Know: How to Perform Vulnerability Prediction using Noisy Historical Data,” May 19, 2023.
- **ASE 2022**, Michigan, USA: Presented “Cerebro: Static Subsuming Mutant Selection,” October 12, 2022.

Publications

1. **Aayush Garg**, Zanis Ali Khan, Renzo Degiovanni, Qiang Tang. “[Evaluating LLMs for One-Shot Patching of Real and Artificial Vulnerabilities.](#)” *ACM/SIGAPP Symposium on Applied Computing (SAC) Smarter Engineering - Building AI and Building with AI (SEAI)*, 2026.
2. Zanis Ali Khan, **Aayush Garg**, Yuejun Guo, Qiang Tang. “[Evaluation of Multi-Language Vulnerability Patching with Pre-Trained Models.](#)” *ACM/SIGAPP Symposium on Applied Computing (SAC) Computer Security (SEC)*, 2026.
3. Zanis Ali Khan, **Aayush Garg**, Qiang Tang. “[A Multi-Dataset Evaluation of Models for Automated Vulnerability Repair.](#)” *International Workshop on Artificial Intelligence, Cyber and Cyber-Physical Security (AI&CCPS), ARES*, 2025.
4. **Aayush Garg**, Renzo Degiovanni, Mike Papadakis, Yves Le Traon. “[On the Coupling between Vulnerabilities and LLM-generated Mutants: A Study on Vul4J dataset.](#)” *IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2024.

5. **Aayush Garg**, Yuejun Guo, Qiang Tang. “[AI-Driven Software Security: Vulnerability Detection, Patching, and Anti-Fuzzing.](#)” *The 139th European Research Consortium for Informatics and Mathematics (ERCIM) News, Special Theme: Software Security*, 2024.
6. **Aayush Garg**, Renzo Degiovanni, Facundo Molina, Mike Papadakis, Nazareno Aguirre, Maxime Cordy, Yves Le Traon. “[Enabling Efficient Assertion Inference.](#)” *IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 2023.
7. Milos Ojdanic, Ahmed Khanfir, **Aayush Garg**, Renzo Degiovanni, Mike Papadakis, Yves Le Traon. “[On Comparing Mutation Testing Tools through Learning-based Mutant Selection.](#)” *ACM/IEEE International Conference on Automation of Software Test (AST)*, 2023.
8. **Aayush Garg**. “[Guiding Quality Assurance Through Context Aware Learning.](#)” *Ph.D. Dissertation, University of Luxembourg Open Repository and Bibliography (ORBilu)*, 2023.
9. Milos Ojdanic, **Aayush Garg**, Ahmed Khanfir, Renzo Degiovanni, Mike Papadakis, Yves Le Traon. “[Syntactic Vs. Semantic similarity of Artificial and Real Faults in Mutation Testing Studies.](#)” *IEEE Transactions on Software Engineering (TSE)*, 2023.
10. **Aayush Garg**, Renzo Degiovanni, Matthieu Jimenez, Maxime Cordy, Mike Papadakis, Yves Le Traon. “[Learning from What We Know: How to Perform Vulnerability Prediction using Noisy Historical Data.](#)” *Empirical Software Engineering (EMSE)*, 2022.
11. **Aayush Garg**, Milos Ojdanic, Renzo Degiovanni, Thierry Titchou Chekam, Mike Papadakis, Yves Le Traon. “[Cerebro: Static Subsuming Mutant Selection.](#)” *IEEE Transactions on Software Engineering (TSE)*, 2021.

Preprints

1. **Aayush Garg**, Abdelwahab Boualouache, Adnan Imeri, Uwe Roth. “[A Survey of Cyber Range Training Exercise Scenario Description Generation and Execution.](#)”
2. **Aayush Garg**, Constantinos Patsakis, Zanis Ali Khan, Qiang Tang. “[Payload Analysis of Adversaries’ Tooling: Automated Identification of Fuzzers.](#)”
3. **Aayush Garg**, Renzo Degiovanni, Mike Papadakis, Yves Le Traon. “[Vulnerability Mimicking Mutants.](#)”
4. **Aayush Garg**, Renzo Degiovanni, Matthieu Jimenez, Maxime Cordy, Mike Papadakis, Yves Le Traon. “[Learning to Predict Vulnerabilities from Vulnerability-Fixes: A Machine Translation Approach.](#)”

References

Available upon request.